TECHNICAL INTERVIEW GUIDE · DEEP DIVE

Senior Backend Engineer

Orders team · 75 min

CANDIDATE | INTERVIEWER | DATE

THE CASE

Our food delivery app takes about 2,000 orders a minute at lunch peak. The order service is written in Python, uses PostgreSQL on AWS and calls an external payment provider. Last month, app retries during a slow payment period created duplicated orders and double charges. In the first months, this person will own the order placement flow and get it ready for twice today's peak.

What this role protects: APIs and data consistency · PostgreSQL under load · Resilience to slow dependencies

HOW TO EVALUATE

C Correct — accurate, no hand-waving	**R** Reasoning — explains why and weighs trade-offs	**E** Edge cases — failure modes, limits, scale	**H** Hands-on — has done it, with real details

Score: 4 expert (all four, holds up when you dig) · 3 solid · 2 basic · 1 gaps
Business questions (marked STAR): Situation, Task, Action, Result

1 · FUNDAMENTALS

1 **An index made our order lookup fast, but inserts slowed down at peak. Explain why, and how you'd decide which indexes to keep.**

Dig deeper: How would you see this in PostgreSQL?
Strong answer: *Write cost of each index, bloat, read vs write ratio; uses query stats and EXPLAIN to decide with data.*

Heard ☐ C ☐ R ☐ E ☐ H Score ◯ 1 ◯ 2 ◯ 3 ◯ 4

CANDIDATE'S ANSWER

2 **Two requests update the same order at the same time. What can go wrong, and how do transaction isolation levels help?**

Dig deeper: When would you lock the row instead of raising the isolation level?
Strong answer: *Lost updates and anomalies per level; row locking vs optimistic locking with a version column.*

Heard ☐ C ☐ R ☐ E ☐ H Score ◯ 1 ◯ 2 ◯ 3 ◯ 4

CANDIDATE'S ANSWER

3 **What does it mean for an endpoint to be idempotent, and how would you make POST /orders idempotent?**

Dig deeper: Where do you store the idempotency key, and for how long?
Strong answer: *Client key, unique constraint, stored response, TTL; handles two identical requests arriving at once.*

Heard ☐ C ☐ R ☐ E ☐ H Score ◯ 1 ◯ 2 ◯ 3 ◯ 4

CANDIDATE'S ANSWER

4 **When would you process something with a queue instead of inside the request, and what do you give up?**

Dig deeper: How do you avoid losing a message if the service crashes right after writing to the database?
Strong answer: *Latency and decoupling vs consistency and complexity; at-least-once delivery, outbox pattern, deduplication.*

Heard ☐ C ☐ R ☐ E ☐ H Score ◯ 1 ◯ 2 ◯ 3 ◯ 4

CANDIDATE'S ANSWER

2 · HANDS-ON AND TROUBLESHOOTING

5 **p95 latency of POST /orders went from 200 ms to 2 s at lunch and the database CPU is at 90%. Walk me through it.**

Dig deeper: What would make you change your first hypothesis?
Strong answer: *Method: metrics, slow queries, locks, connections; narrows down with data; mitigates before the root cause.*

Heard ☐ C ☐ R ☐ E ☐ H Score ◯ 1 ◯ 2 ◯ 3 ◯ 4

CANDIDATE'S ANSWER

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6 **Tell me about the hardest production bug you've chased. How did you find it?**

Dig deeper: What did you try first that didn't work?

Strong answer: *Real case, hypotheses tested with data, reproduced it, verified the fix and added a test or an alert.*

Heard ☐ C ☐ R ☐ E ☐ H Score ◯ 1 ◯ 2 ◯ 3 ◯ 4

CANDIDATE'S ANSWER

7 **We see duplicated orders a few times a day. How would you find where they come from?**

Dig deeper: How would you prove your fix worked?

Strong answer: *Correlates request IDs and logs, looks for retries and races, checks constraints; measures duplicates after.*

Heard ☐ C ☐ R ☐ E ☐ H Score ◯ 1 ◯ 2 ◯ 3 ◯ 4

CANDIDATE'S ANSWER

8 **Tell me about a time a deploy of yours broke production. What did you do, and what changed after?**

Dig deeper: How long until customers stopped being affected?

Strong answer: *Fast rollback, clear communication, blameless follow-up with concrete prevention (tests, canary, flags).*

Heard ☐ C ☐ R ☐ E ☐ H Score ◯ 1 ◯ 2 ◯ 3 ◯ 4

CANDIDATE'S ANSWER

3 · ARCHITECTURE AND DESIGN

9 **Design the order placement flow for 2,000 orders a minute at peak, so a retry from the app or a slow payment provider never creates a duplicate order or a double charge. Draw it end to end.**

Dig deeper: Walk me through what happens if the payment provider times out after charging.

Strong answer: *Idempotency keys, order state machine, outbox or queue, reconciliation with the provider; covers each failure path.*

Heard ☐ C ☐ R ☐ E ☐ H Score ◯ 1 ◯ 2 ◯ 3 ◯ 4

CANDIDATE'S ANSWER

10 **How would you model the orders and payments tables for your design?**

Dig deeper: Which database constraints protect you even if the code has a bug?

Strong answer: *Clear states, unique keys for idempotency and provider IDs, foreign keys, audit trail; money stays consistent.*

Heard ☐ C ☐ R ☐ E ☐ H Score ◯ 1 ◯ 2 ◯ 3 ◯ 4

CANDIDATE'S ANSWER

11 **In your design, what happens when the payment provider is down for 10 minutes at peak?**

Dig deeper: What does the customer see in the app?

Strong answer: *Timeouts, circuit breaker, pending state or queue, clear message to the user, retry and reconcile later.*

Heard ☐ C ☐ R ☐ E ☐ H Score ◯ 1 ◯ 2 ◯ 3 ◯ 4

CANDIDATE'S ANSWER

12 **If you had half the time to ship, what would you cut from your design, and what would you never cut?**

Dig deeper: What risk are you accepting with that cut?

Strong answer: *Keeps idempotency and money consistency; defers nice-to-haves; names the accepted risk explicitly.*

Heard ☐ C ☐ R ☐ E ☐ H Score ◯ 1 ◯ 2 ◯ 3 ◯ 4

CANDIDATE'S ANSWER

4 · SCALE, RELIABILITY AND SECURITY

13 **Traffic doubles next year. What breaks first in your design, and how would you know before it does?**

Dig deeper: Which load test would you run, and what result means we're ready?

Strong answer: *Finds the bottleneck (writes, connections), load tests with a clear threshold, metrics and alerts, scaling steps.*

Heard ☐ C ☐ R ☐ E ☐ H Score ◯ 1 ◯ 2 ◯ 3 ◯ 4

CANDIDATE'S ANSWER

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14 **How would you scale PostgreSQL writes for orders without losing consistency?**

Dig deeper: When would you split the database, and what would it cost?
Strong answer: *Connection pooling, query tuning, partitioning, replicas for reads; sharding only with a clear need and its costs.*

Heard ☐ C ☐ R ☐ E ☐ H Score ◯ 1 ◯ 2 ◯ 3 ◯ 4

CANDIDATE'S ANSWER

15 **Which security risks worry you most in an order and payment API, and how would you handle them?**

Dig deeper: How do you keep card data out of your systems?
Strong answer: *AuthN and AuthZ, validation, rate limiting, secrets, tokens from the provider instead of card data, audit logs.*

Heard ☐ C ☐ R ☐ E ☐ H Score ◯ 1 ◯ 2 ◯ 3 ◯ 4

CANDIDATE'S ANSWER

16 **Which metrics and alerts would you put on the order flow to catch problems before customers do?**

Dig deeper: Which one would wake you up at night?
Strong answer: *Order and payment success rate, latency, duplicates, queue age; alerts on customer impact, not CPU.*

Heard ☐ C ☐ R ☐ E ☐ H Score ◯ 1 ◯ 2 ◯ 3 ◯ 4

CANDIDATE'S ANSWER

5 · BUSINESS AND TEAMWORK

17 **Tell me about a technical decision of yours that saved or cost the business money. How do you know?**

Dig deeper: What number did you or someone else measure?
Strong answer: *Real decision, impact in money, customers or time with a number, and what they learned.*

Heard ☐ S ☐ T ☐ A ☐ R Score ◯ 1 ◯ 2 ◯ 3 ◯ 4

CANDIDATE'S ANSWER

18 **Product wants a new promotions feature next month, but you want to fix the duplicated orders first. How do you make the case?**

Dig deeper: What would you accept as a compromise?
Strong answer: *Explains the cost of duplicates (refunds, support, trust), proposes sequencing or scope, decides together.*

Heard ☐ S ☐ T ☐ A ☐ R Score ◯ 1 ◯ 2 ◯ 3 ◯ 4

CANDIDATE'S ANSWER

19 **Tell me about a time you explained a technical trade-off to someone non-technical. How did it go?**

Dig deeper: How did you know they understood?
Strong answer: *Plain words, focus on impact and options, checks understanding; a real outcome.*

Heard ☐ S ☐ T ☐ A ☐ R Score ◯ 1 ◯ 2 ◯ 3 ◯ 4

CANDIDATE'S ANSWER

20 **How would you explain the duplicated orders to the support team in two sentences?**

Dig deeper: What would you ask them to tell customers?
Strong answer: *Plain words: what happened, who's affected, what's being done and when. No jargon, no blame.*

Heard ☐ S ☐ T ☐ A ☐ R Score ◯ 1 ◯ 2 ◯ 3 ◯ 4

CANDIDATE'S ANSWER

FINAL ASSESSMENT

STRENGTHS

CONCERNS

AVERAGE SCORE (1-4)

RECOMMENDATION

◯ Strong yes ◯ Yes ◯ No ◯ Strong no

NEXT STEP

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